# Honors Portfolio Reflection Essay

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# Intro

In this reflection, I will discuss 11 important artifacts which shaped the course of my academic career at NCSU.  For each, I will provide an overview of what the artifact is, my process with creating / interacting with it, and my key takeaways, and how it connects to other areas.

Here is a list of the artifacts, in the order they are discussed:

### Research

1. Math Minor (Academic)
2. MA 326 adopt an algorithm paper (Academic)
3. MA 404 adopt an algorithm presentation (Academic)
4. RISE REU research + poster (Academic)

### Design

5. E 101 FEDD Project (Academic)
6. Aluminum Coaster (Academic)

### Metaskill

7. Metaskill Challenge (Non-Academic)

### Hobbies

8. Origami (Non-Academic)
9. Yarn (Climate Change Scarf) (Non-Academic)
10. Climbing Gear (Non-Academic)
11. Wind ensemble Costa Rica Trip (Non-Academic)

# Research (Intro and Practice)

## Math Minor

One of the most pivotal experiences in my undergraduate experience stemmed from a single seemingly inconsequential decision. As part of the UHP requirements, I am required to take 6 credits of *honors approved coursework*. One course that caught my eyes was an honors section of *MA 401: Differential Equations 2*. I have enjoyed math as long as I have been in school, and much prefer it to anything with an essay, so this course was an obvious yes. With this course and its prerequisite *MA 341: Ordinary differential equations*, I realized I was already 40% of my way to completing a minor in mathematics (In my second semester, at that)! As an extra bonus, for a *College of Engineering* student such as myself, a math minor is eligible to count as a HIE, due to disciplinary breadth (as it is a minor in a different department).

With this realization, I was emboldened to go all-out and commit to completing the math minor. For some context, all the classes up to *Calculus 3 (MA 241)* are required as a prerequisite for a math minor. After that, you must complete 15 more credits of 300/400 level Math classes (5 classes, typically). The next class I took on this path was *MA 405: Linear Algebra*. Though it is unassuming, knowledge of this topic opens up an incredible number of doors in an academic setting. There are multiple hard classes required for my degree (*ECE 331* is a notorious one) where a firm grasp of linear algebra puts me miles ahead of peers. Additionally, it laid the groundwork for me to take several advanced classes in the future (I am currently taking a 500-level Operations Research class built out of linear algebra).

The last two classes I took towards my mathematics minor come as a pair, and they constitute two of the trickiest, yet most interesting classes I have taken in my academic career. The two *Mathematical Foundations of Data Science* courses, *MA 326* and *MA 404*, are a deep-dive into the motivations and inner workings of numerous modern data science algorithms. As a small taste of what this entails, we move all the way from linear regression and logistic regression (the foundational calculation occurring at each node of a neural network) at the beginning of MA 326, to characterizing the transformers, attention mechanisms, and other components which power ChatGPT 2 at the end of MA 404.

# MA 326: Adopt an Algorithm Paper

With their depth, these two classes are both quite research-oriented. To further encourage this, both courses include a capstone: the “Adopt an Algorithm” Project. The aim of this project is simple: choose an interesting data science algorithm, characterize its behavior and math behind it, demonstrate the algorithm in action, and communicate your results.

For the first class (MA 326), I choose to dive into *decision trees*. In the current era, most of the discussion we hear about data science is focused on *Large Language Models* and other models derived from Neural Networks (NNs). Before these NNs rose to dominance, decision trees (and applications thereof) were seen as the gold-standard for processing structured data, and they still outperform NNs in many applications.

This was a very fun project for me, as it was my first chance in quite a while to really get my hands dirty and use my coding skills to put together something complicated. The biggest challenge I faced was having to figure out *Object-Oriented Programming (OOP)* in python. Decision trees work by recursively splitting their input data (forming a tree), which requires generation of a potentially arbitrary collection of nodes and connections (OOP is perfect here, as it allows creation of numerous instances of a base object with different characteristics/data). I enjoyed the coding part of this project so much, in fact, that I decided to expand my model to generate an ensemble of decision trees, aka a *random decision forest*. A given dataset is split and sampled to create several smaller, independent datasets which are then fed to a multitude of decision trees. This method can create a very powerful predictive model, as the large quantity of independent predictions tend to converge around a correct result better than a single estimate (The data science equivalent of asking several opinions for something instead of just one).

After implementing and testing my algorithm, the project culminated in a ten page research paper. I definitely struggled a bit to put the pieces together, but this was understandable for my first attempt at a technical research paper (as an additional painful note I wrote the entire paper, formulas included, in google docs…). This paper was a large milestone for me, as it represented my first big step into research and academia.

## MA 404: Adopt an Algorithm Presentation

The second semester of the data science class (MA 404) was a peculiar semester; it was the first semester it was offered, and there were only seven people in the class (including the teacher)! The project took a slightly different form this semester, as the small class size allowed us to structure our communication as a 30 minute presentation. To keep up with the theme of the class (optimization algorithms and NNs), I choose to research the *Differential Evolution* (DE) method for optimization.

Minimizing a loss function is a very common topic in data science, and DE is an approach to this end which borrows ideas from the field of biology. To put it briefly, DE creates a cloud of points, each with a set of “genes” (the coordinates of the points). In a process which mimics natural selection, it takes points with favorable loss values (high “fitness”) and combines and mutates them in an attempt to achieve children with better loss values. The resulting behavior of this algorithm is a swarm of points which gradually hone in on loss minimizers, while remaining scattered enough to avoid getting stuck in poor local minima.

The presentation of these results posed an interesting challenge, due to the differing format from the previous project. As I was creating a presentation (not a paper), I needed to carefully ensure that my descriptions weren’t too dense (the prior paper was very dense). To achieve this end, I leaned into using python to generate visualizations of the training process, where you could watch the cloud of points close in on an optimal solution to a small problem. I got a lot of good practice with my visualization skills implementing this, from graphing the loss functions, placing the DE points, and combining frames from each iteration to produce a convenient GIF. Lastly, this project gave me much needed experience presenting, as it marked the longest solo research presentation I had given.

## RISE REU Research and Poster

These “Adopt an Algorithm” papers were such a good basis for research, that the first one actually led to me getting a research internship via the *RISE Research Experience for Undergraduates* (REU) program (starting Summer 2025)! My project was on combining the strengths of agent-based and compartmental (differential-equation based) models, using a

*Biologically-Influenced Neural Networks (BINNs)* technique. I focused specifically on building off of a previous researcher's application of the BINNs technique to model the behavior of an Epidemiological model called *Covasim*. The goal of my research was to flesh out the base compartmental model used previously, to improve biological plausibility, and hopefully improve the predictions of the model as well.

This summer program culminated in my first research poster, and a presentation at NCSU's *Summer Undergraduate Research Symposium*. I have since continued work on this research to this semester. I have gained many valuable skills from this project, including significantly improved python skills (more OOP, neural network architecture setup and training, and more), knowledge of GitHub for code sharing and version control, and experience reading, writing, and interpreting complex research papers.

# Academic Progression

## E 101: FEDD Project

Continuing my discussion of academic projects, I will bring up a pair of major design projects I have worked on during my undergraduate degree, and what I have learned. The first of these projects, and a canon event for all NCSU Engineering students, is the *Freshman Engineering Design Day* (FEDD) project. It is simple at its core, with students forming teams around one of a standard set of projects, but a secondary purpose lurks under the surface. Many freshmen do not have particularly good communication/teamwork skills, time management skills, nor knowledge of all the resources available. If teams cannot work around these issues, their projects will inevitably fall to chaos.

My group chose the *Mechanical Music Maker* project, with the goal of creating a mostly (>50%) mechanical device which can play a short, recognizable tune when turned on. We ended up designing a small, handheld music box which would play the song *Minecraft* by C418 when the crank was turned. Due to a misunderstanding on my part, we ended up having to modify the comb of an existing music box with a dremel to get the correct notes (we were not allowed the tools required to manufacture our own). Due to poor time management, we had to make the

comb-plucking device out of a section of pool noodle with toothpicks stuck through it; this fixture being integrated with the comb via lego technic pieces, hot glue, and a dream.

When FEDD day came, we presented, and the music box did not work very well. It made sounds, but they were not pleasant; many of the notes were either too loud or completely missed (based on toothpick placement). Despite these issues, we somehow ended up in *3rd place out of 10* groups. You might be wondering, how did this happen? The aforementioned chaos came to our advantage, and most of the groups didn't have a music maker which could make any kind of music. As a result, there were only three viable competitors, with one of them being disqualified due to "not being mechanical enough." Additionally, although our music box sounded completely wrong, none of the judges actually knew the song we wanted it to play, so they had no way of knowing that! In the end, our group went home with free T-shirts to commemorate our 3rd place ranking. In the end, we had many important takeaways: working as a team is very important, as is being proactive. And even when things go wrong, confidence is key, and it can help you get out of some sticky situations.

## ISE 315/316: Aluminum Coaster

The next major design project I completed was the *Aluminum Coaster* project for my ISE 316 class. This project was a big step up from the FEDD project, as we needed to design an NCSU-themed aluminum coaster, making several important compromises along the way, then actually manufacture a set of them! To top things off, we needed to write a report with detailed documentation of our design and manufacturing process, along with a plan and cost estimate to scale production up to meet a demand of 100,000 units.

Despite its complexity, this project ran very smoothly. This performance is due to a combination of better time management, efficient division of responsibilities, and better team cohesion due to common goals. In addition to demonstrating these improvements, I also got to flex my design muscles more with this project (I was the lead concept + CAD designer), and get more practice with my report-writing. Lastly, since our group made a full set of coasters, we all got to take one home as a souvenir!

# Metaskill Challenge

Something interesting I have noticed about myself is that I tend to thrive in environments where I get to teach myself at my own pace;  this quality has been quite advantageous both in and out of the classroom.  With this in mind, it is no wonder that the *20 Hour Metaskill Challenge* has been one of the highlights of my college experience.

I have an older brother who is very artistically inclined, so I thought it would be fun to try out something similar for this challenge.  With this motivation, and the fact I have a computer which can fold into a tablet + stylus, I figured I could pursue digital art.  Thus, I set out on a journey planning to explore the brushes and tools for digital art (in Krita), while teaching myself basic drawing and construction techniques.  These techniques should be sufficient to allow me to break whatever I see down into basic shapes, letting me draw them piece-by-piece.

Although it was well-intentioned, my plan fell apart pretty fast, due to one critical error.  Digital art as a medium is very good for refining things, which means it is also very easy to spend a lot of extra time if you are not sure of your skills.  Secondly, the lack of friction on my laptop screen made it incredibly difficult to draw well, due to my shaky hands (trying to draw on a frictionless surface severely hinders your muscle memory if you have only ever written on paper).  With these two drawbacks of digital art, I ended up burning out relatively quickly.

To rectify these issues, I switched primarily to pen and paper.  I also took up various drawing exercises, to help gain confidence and fix my shaky hands.  After weeks of practice in this manner, I was able to return to my laptop.  The challenge culminated in a 5 minute video, including a scenic speedpaint which I completed using my laptop.  In the end, I was able to come to a result I liked, regardless of the setbacks.  In addition to the clear improvements to my art skills, I also learned some basic video editing skills, which I have gone on to utilize for some personal projects.  I have also continued drawing since the end of the challenge.  Interestingly enough, completing the challenge felt freeing, in a way, as it led to me realizing I enjoy drawing more when I draw for myself, instead of drawing to share with others.

# Hobbies

Next, we can change the pace a bit and talk about some less serious artifacts. I often feel at my best when I am working on something I enjoy, whether it is a small, intricate work of art or a vigorous, full-body movement puzzle. Thus, the remaining focus will be on some of the hobbies I have juggled through my college experience.

## Origami

I was introduced to origami by my fourth grade teacher, and I have been hooked ever since. When one folds so many things over more than a decade, some interesting problems expose themselves: You start running out of room to keep everything. For the pieces you choose to keep, the creases in the paper will naturally loosen over time. In extreme cases, this can lead to complete structural collapse in complex geometric (several interlocking units) models. These factors lend an intriguing degree of transience to the art of origami as a whole, which has shaped my view of it over time. In previous years, I was obsessed with crafting several highly intricate pieces – some with multiple dozens of pieces – to showcase (Attach image). In the last few years, however, I have often found more enjoyment in slowing down and appreciating the process itself, along with the value of letting go. Over the last few semesters, I have started curating small collections of origami in the common areas of my dorm. It is cool to watch pieces slowly accumulate as time passes, until the slate is wiped clean for the start of the next semester (thank you, housekeeping staff!). (Attach image)

## Yarn

Another pastime I have enjoyed for the last few years can be a bit more of a tangled mess, but it is still very fun. I learned to crochet around 4 years ago, and to knit more recently. Some may see these fiber arts as nothing more than a medium to produce scarves, hats, stuffed animals, and whatnot (and I have made my fair share of these), but there is so much more you can do with them.

A fairly common project idea for crocheters is a temperature blanket: each day, you add a row to your blanket, with a color determined by the temperature that day. An interesting prompt

from my Honors Seminar *IS 200: Intro to International Studies* led me to try a wild twist on this idea, in the form of a "climate change scarf." Drawing upon a report from the *Intergovernmental Panel on Climate Change*, I compiled historical data and future predictions for atmospheric temperatures, spanning from 1900-2100. This information was used to design a scarf with two parts: a fixed block representing the past, plus five long ribbons representing the future environment under different control measures. The final result was a striking, wearable piece which showcases the importance of taking care of our world.

## Climbing

Speaking of taking care of the world, my next point is actually an example of how the UHP has helped me to take care of myself. During my first semester at state, I signed up for a forum event at the *OA Climbing Center*, thinking it would be a fun one-time event… I was quite wrong. Instead, I struggled spectacularly, and had a ton of fun doing it. The full-body workout, along with the active, hands-on problem solving required, came together as the first form of exercise I have legitimately enjoyed (some people would even consider it addictive). This activity has led to a noticeable improvement in my health, as I have lost at least 20lbs as a direct result, and gotten much stronger (enough to complete 10 pull-ups on a good day!). I have also improved massively at climbing, going from struggling on V0-V1s and 5.5s (curse you, PUMPkin Patch!) to completing crazy projects up to V7 and 5.12 in the last year. I have also participated in numerous events, completing both the classic *Faux Rock 50* and the recently introduced *Top-rope Happy Hour* challenge (no alcohol included). I have also competed in the *Boulder Bash* and the *Top-rope Throwdown*, placing and receiving prizes on multiple occasions.

## Music and Costa Rica

One last hobby that has taken me places I never expected is my passion for music. I have played the french horn since 6th grade, the piano since before that, and many other instruments. To keep improving my skills, I have played french horn with NCSU's Wind Ensemble since arriving here. The summer before last (2025), we got an opportunity to go perform a series of concerts in Costa Rica, with a main performance in *El Teatro Nacional*. As one would expect, we spent the majority of the trip exploring many of the country's attractions, and had an absolute blast. We saw monkeys, crocodiles, caimans, birds, and more along the rivers and jungles,

zip-lined down the side of a volcano, explored a giant water park, toured the flagship Starbucks coffee farm, and much more.  Not to forget the food we tried there; it was absolutely amazing! All of the fruit tasted incredibly good, and one of the restaurants I stopped at with a smaller group had the best tacos I have ever tasted.  With this trip, I would have to say Costa Rica is the best travel destination I have ever visited.

# Conclusion

Looking back on everything, I definitely enjoyed participating in the UHP, and I got a lot out of it. Some opportunities, such as the honors section of MA 401, ended up completely shifting the course of my academic career.  Others things, such as the climbing clinic, changed my life for the better in a measurable way.  I would recommend the UHP for students who would like to get the most out of all of their opportunities;  the honors forum and other coursework can open up so many doors.